

Sample, data collection and instruments

Quantitative designs

Surveys

Is the **purpose** of the survey design clearly specified?

What **type of design will be used**, and what are the reasons for choosing

Is the **nature of the survey** identified

Is the **population described** and is its size indicated?

Will the **population be stratified**? If so, how will stratification be implemented?

How many individuals will be in the sample, and on what basis was this size chosen?

What will be **the procedure for sampling** these individuals

What **instruments** will be used in the survey

Do the instruments have acceptable score reliability and validity, and what are the scale anchors?

What is **the timeline** for administering the survey?

What **ethical considerations** will be taken into account to protect participants?

1

Purpose of the instrument

2

Sampling process

3

Instrument development

4

Instrument administration

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Surveys

1

**Purpose of the
Instrument**

2

Sampling
Process

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The purpose

To answer one or more **research questions about variables of interest.**

- ✓ *“The main purpose of this survey study is to empirically assess whether the number of overtime hours worked predicts subsequent symptoms of burnout in a sample of emergency-room nurses.”*

The justification

When justifying the choice of method, it is essential to **recognise the advantages of survey designs:**

- Cost-effectiveness
- Rapid data collection
- Constraints that prevent the use of other designs
- ✓ *“An experimental design was not adopted to observe the relationship between overtime hours and symptoms of burnout because it would be prohibitively costly and potentially unethical to randomly assign nurses different amounts of overtime.”*



Fowler (2014)

Type of Survey Method

Indicate whether the survey will be **cross-sectional**, with data collected at one point in time, or **longitudinal**, with **data collected over time.**

Modes of Data Collection

Fowler (2014) identifies the following types:

- ✓ Mail
- ✓ Telephone
- ✓ Face-to-face interviews
- ✓ Internet
- ✓ Group administration

Regarding internet surveys and online administration, there has been significant growth in their popularity because they provide easy access to participants around the world, such as through platforms like Amazon Mechanical Turk and Prolific.

Regardless of the data-collection mode, it is important to justify the chosen procedure, drawing on its strengths, weaknesses, costs, data availability, and convenience.

Quantitative designs

Surveys

Key approaches to obtain factors that would otherwise remain unobservable:

- ✓ Perceptions
- ✓ Knowledge
- ✓ Attitudes
- ✓ Beliefs

These factors are important because **they are determinants of political, social, and economic outcomes.**

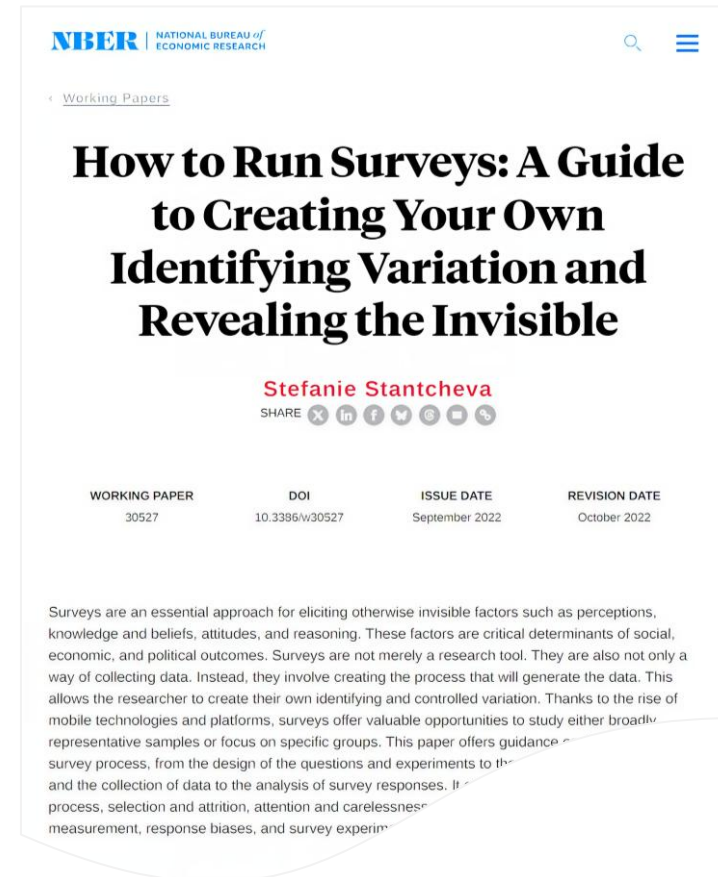
We are designing and building a process that enables the **generation of data.**

- ✓ This allows us to control and identify the level of variation in the information we produce.
- ✓ Survey design is an integral part of the research process.
- ✗ ! Surveys should be understood only as one method of data collection.



Stefanie Stantcheva

Social economy Lab
at Harvard



Quantitative designs

Surveys

A clear definition of the research question

A good survey must be adapted to the subject of the research.

- When writing survey questions, we must always think about **how we are going to analyse them**
- This means that the design of the survey will **always depend on the objectives established for the research**

In social, economic, and political research settings, **the objective is not to document or collect statistics** (as is the case with administrative records or household surveys).

When designing questions, one must:

- ✓ Keep in mind the concept of ceteris paribus, or “all else held constant”.
- ✓ Consider the exercise as a way of creating controlled variation.
- ✓ Ensure that each question addresses only one aspect at a time, keeping everything else as constant as possible (and respondents should be made aware of this).

Quantitative designs

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Purpose of the
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Sampling

What type of sample is needed to answer the research question?



Sampling Range

The sample must be intentionally diverse in terms of variables that are conceptually relevant to the research question.

These rely on quota-based online panels, such as [Qualtrics](#), [Dynata](#), [Bilendi](#) and [Prolific Academic](#).

Like commercial survey companies, but requiring more hands-on management by the researcher. E.g., [Cint](#).

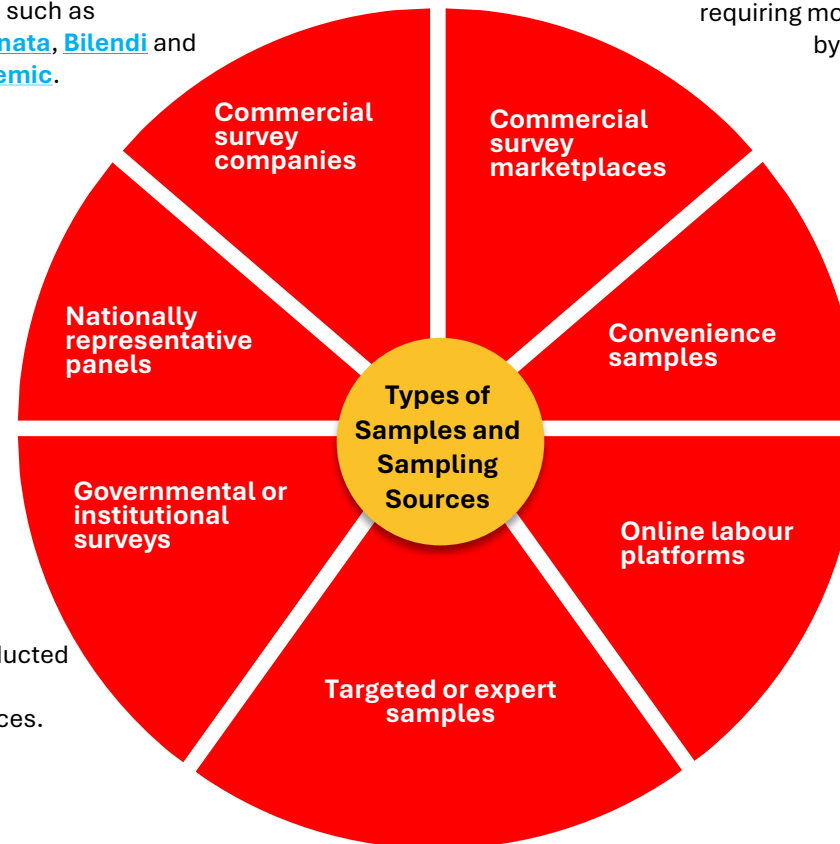
Large-scale national probability panels

Populations that are easily accessible to the researcher, such as university students or conference participants.

Typically conducted by national statistical offices.

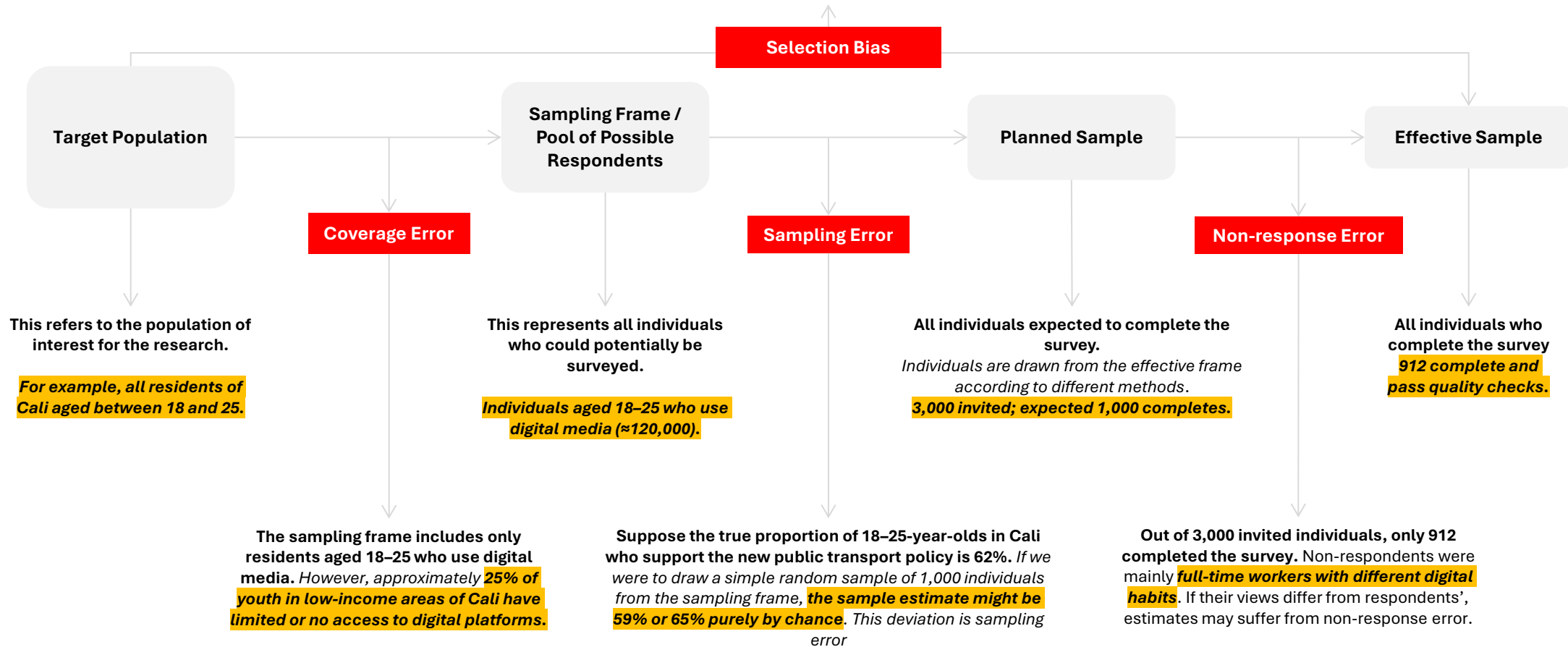
[Such as Mechanical Turk de Amazon \(MTurk\)](#), positioned between convenience samples and quota-based panels due to the large number of available respondents..

For example, experts, employees in a specific firm, economists, or other specialised groups.



Quantitative designs

The systematic difference between the target population and the effective sample, arising from non-random mechanisms at any stage (coverage, recruitment, or response), such that your estimates would be different even with infinite sample size



Quantitative designs

Surveys

Sampling

What risks exist regarding representativeness?

1. Unit Non-response Bias



The difference between the individuals who **actually started the survey and those who were initially expected to do so** (selection-related response error).

2. Item Non-response Bias



Occurs when respondents begin the survey (i.e., are part of the effective sample) **but do not answer all questions**.

✖ Attrition

Refers to the phenomenon whereby respondents **exit the survey before completing it**. This implies that items after a certain point are not fully answered.

If this “drop-off” is systematic rather than random, attrition may introduce bias into the instrument.

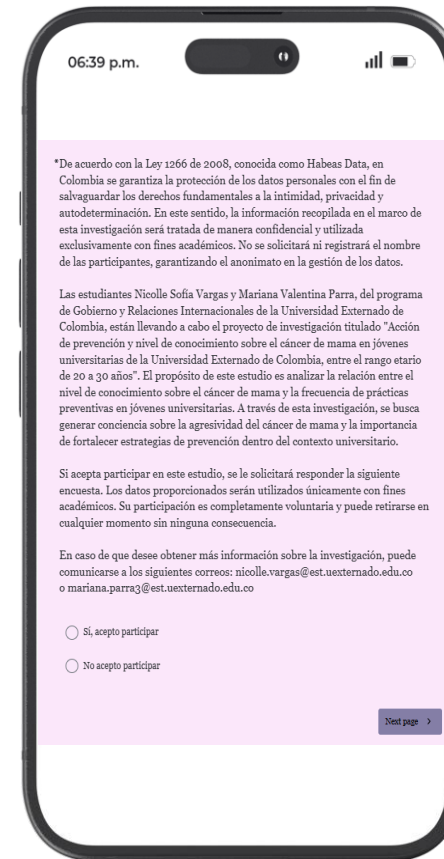
Quantitative designs

Surveys

Recruitment

The First Page of the Survey

- ✓ **Reduce** the perceived costs of participating in the survey from the outset by specifying the expected duration.
- ✓ **Use** simple language and a visually friendly design.
- ✓ **Do not disclose** too much information about the identity of the survey administrator.
- ✓ **Provide** contact information so that respondents may submit complaints, questions, or comments.
- ✓ **Offer** details on how the data will be stored and used.
- ✓ **Reaffirm** the commitment to anonymity and confidentiality.
- ✓ **Present** limited information regarding the purpose/objective of the study.
- ✓ **Explain** the potential benefits derived from participation.
- ✓ **Warn** respondents about the importance of providing high-quality responses.



Quantitative designs

Surveys

Provides a quantitative description of trends, attitudes, and opinions, and tests associations between variables of a population by studying a sample.

1

Descriptive questions

What percentage of the company's employees support the hiring of foreign workers?

2

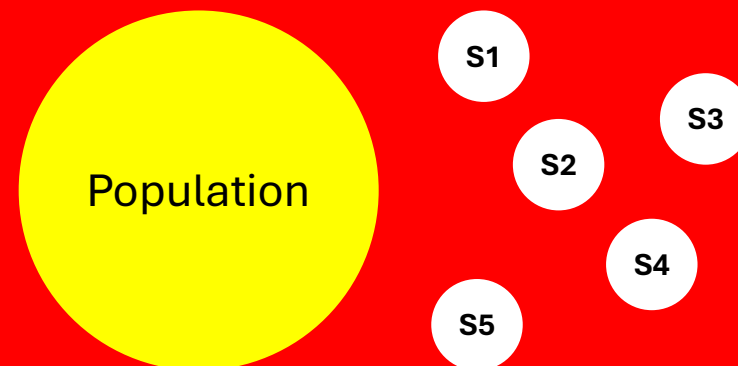
Questions about relationships between variables

Is there an association between the number of foreign employees and overall job-satisfaction perceptions among employees?

3

Longitudinal structures

Does the hiring of new foreign employees at Time 1 predict higher job satisfaction at Time 2?



Quantitative designs

Population

A population is the set of all elements of interest in a study.

Sample

A sample is a subset of the population.

Parameter

A parameter is a fixed characteristic of the population.

Indicators

Indicators (or sample statistics) are quantities computed from the sample.

The primary purpose of statistical inference

The principal goal of statistical inference is to **make estimates and conduct hypothesis tests about population parameters using the information provided by a sample.**

- ✓ It is often difficult, and in many cases costly, to obtain population-level data. Consider the implementation of censuses to gather sociodemographic information.
- ✓ A useful alternative to obtaining population data is to study samples (specifically, their statistics) to infer the characteristics of the population to which the sample belongs.

Quantitative designs

Uncertainty Generated by Random Sampling

Objective

To reproduce the same mean for the entire population, that is, if we want to study the mean, we aim to estimate the average evaluated across all individuals in the population of interest.

1. Expected Value and samples

For any given population, there exists a single expected value $E[Y_i]$.

In contrast, there are many possible sample means $AVG[Y_i]$ depending on how the sample size n is chosen and which individuals happen to be included in the sample.

$$\text{Expected Value} = E[Y_i]$$

$$\text{Sample Mean} = Avg[Y_i] = \bar{Y}$$

2. Population parameters

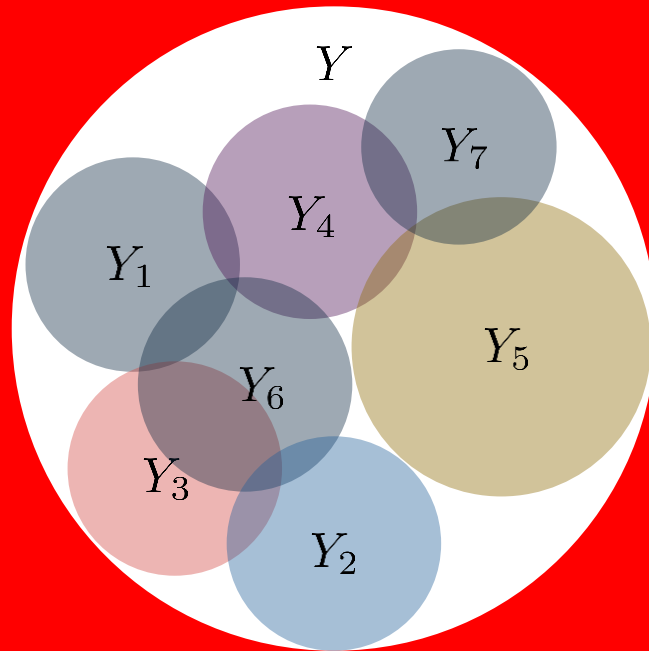
Since $E[Y_i]$ is a fixed characteristic of each population, we refer to it as a parameter.

The quantities that vary from one sample to another are called sample statistics (indicators).

Quantitative designs

Measure	Sample indicators	Population parameters
Mean	$\bar{X}$	μ
Standard deviation	s	σ
Variance	s^2	σ^2
Observations	n	N
Variation coefficient	CV	CV

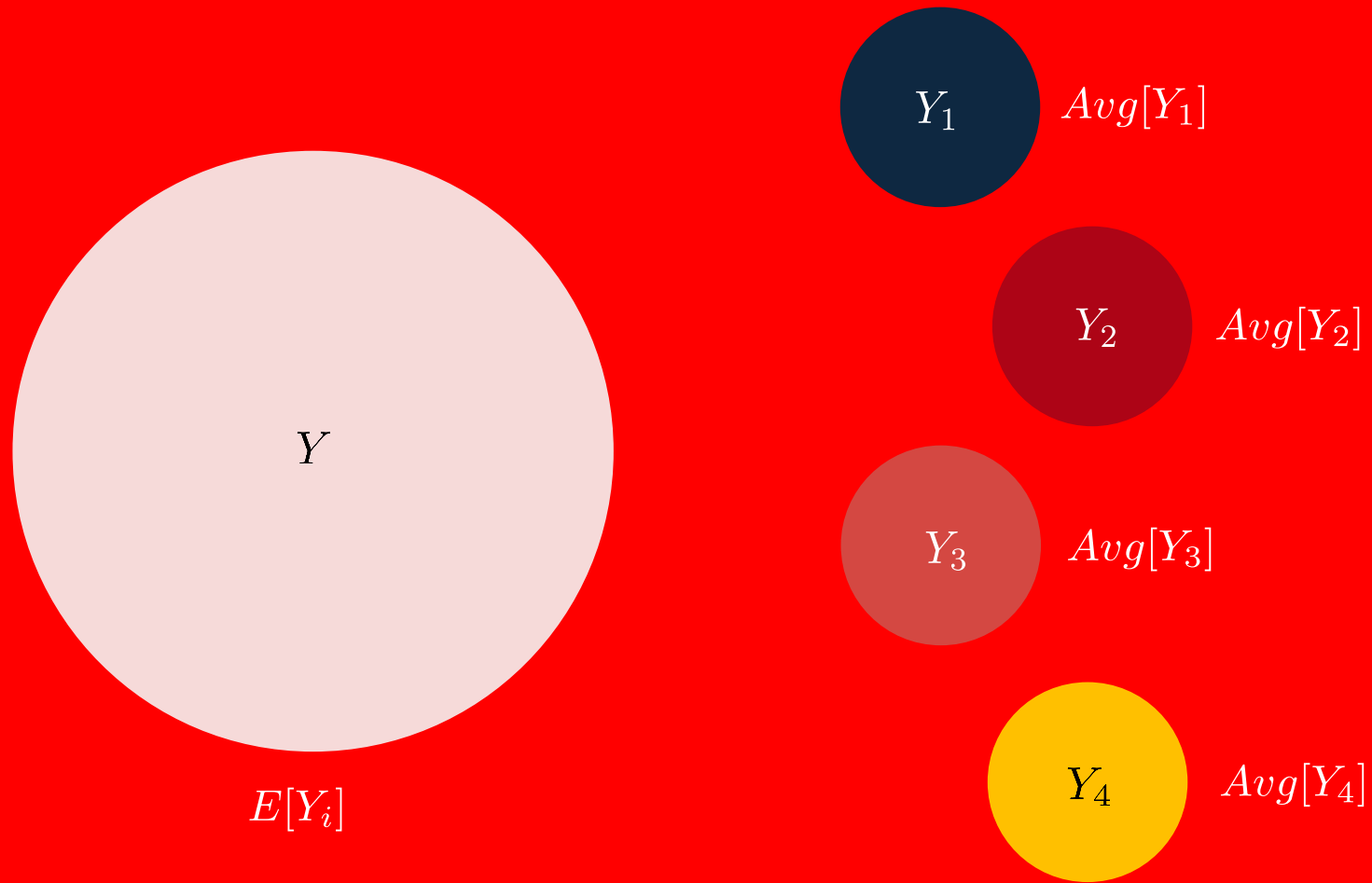
Quantitative designs



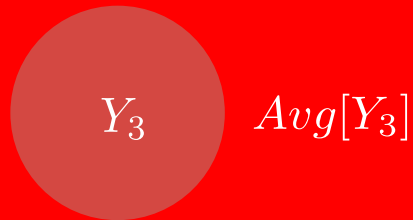
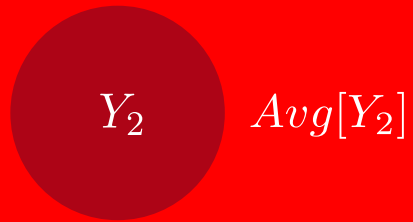
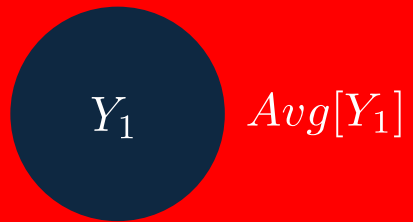
How can we determine **whether statistical results constitute solid evidence**, or whether they are little more than outcomes unlikely to be replicated in another sample?

What level of variability should we expect across samples?

Quantitative designs



Quantitative designs



The law of large numbers tells us that, for large samples, the sample mean is very likely to be very similar to the population mean.

In other words, **if we were to draw infinitely many random samples and compute the sample mean $Avg[Y]$ for each of them, the average of all these sample means would converge to the true population mean, $E[Y]$.**

$$E[Y_i] = E[Avg[Y_i]]$$

A related property states that the expected value of the sample mean, $Avg[Y]$, is equal to the population expected value, $E[Y]$.

Quantitative designs

Uncertainty Generated by Random Sampling

What statistical tests should be used?

A statistical test is a rule defined a priori (i.e., before accessing the data) that specifies **what must be observed in the data to justify rejecting the null hypothesis (H_0)**.

A result meeting this criterion is labelled “statistically significant.”

What is required for a statistical test?

- ✓ **A pre-specified significance level** establishes a “reasonable level of doubt.” **Alpha** is the upper bound on the probability of rejecting H_0 when H_0 is true (Type I error).
 - **For a one-sided test (e.g., $H_0: A \geq B$)**, it is the maximum probability of rejecting H_0 in any situation where $A \leq B$.
 - **For a two-sided test ($H_0: A = B$)**, it is the probability of rejecting H_0 when $A = B$.

Technically, the significance level may be set at any value, provided it is chosen before examining the data.

It is not acceptable to set $\alpha = 5\%$ a priori and then, after seeing the results, change it to 7% post hoc just to reach significance.

- ✗ ! You cannot change the rules of the game once you have seen the outcome.

Quantitative designs

	H0 True	H0 False
Reject H0	✗ Type I error	✓ Correct decision
Do not reject H0	✓ Correct decision	✗ Type II error

The general procedure for hypothesis testing consists of setting the significance level **α at a small value** (often 0.05) so that the probability of committing a **Type I error is limited**.

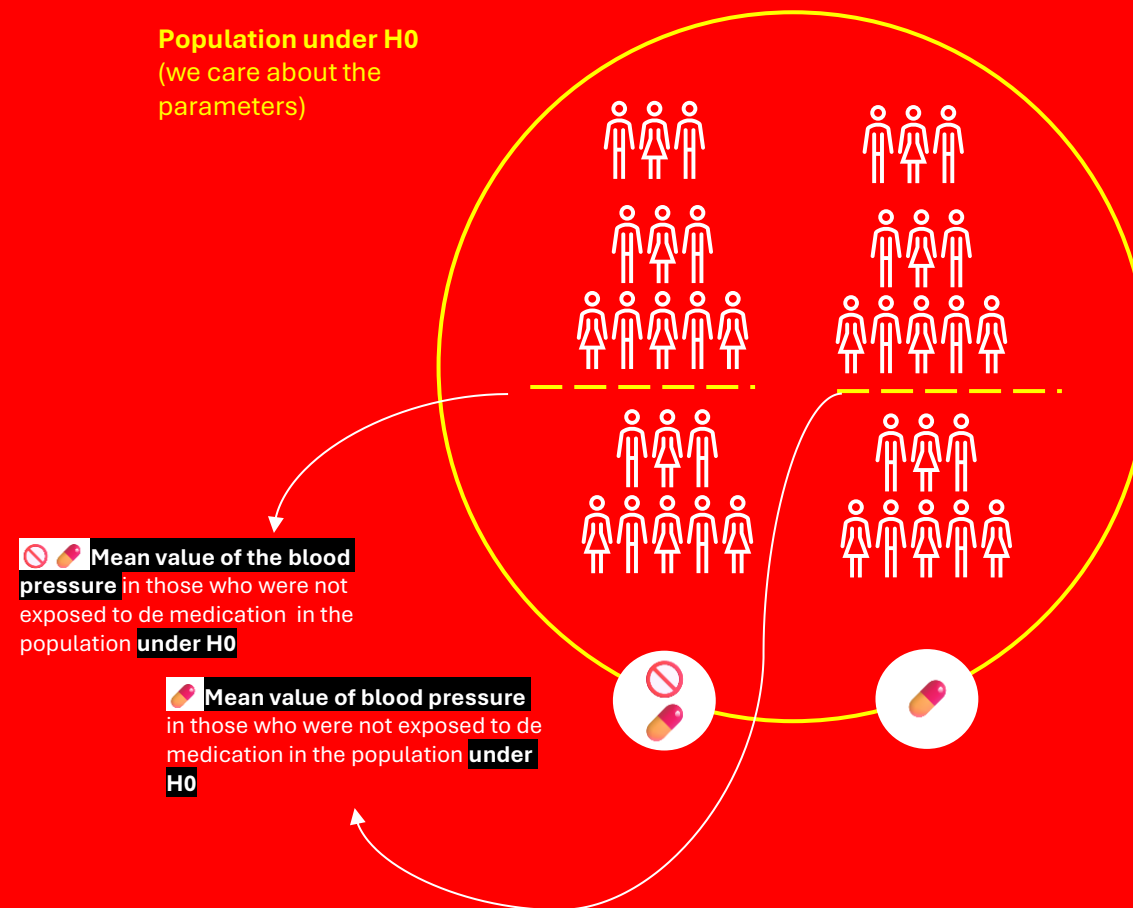
By doing so, we prioritise H_0 over H_1 , acknowledging that a Type I error is typically considered more serious than a Type II error.



Given a fixed value of α , we select a statistical test, defined by its rejection region, that makes β (the probability of a Type II error) as small as possible.

In other words, we aim to find the most powerful test for the specified significance level.

Quantitative designs



Null hypothesis (H0)

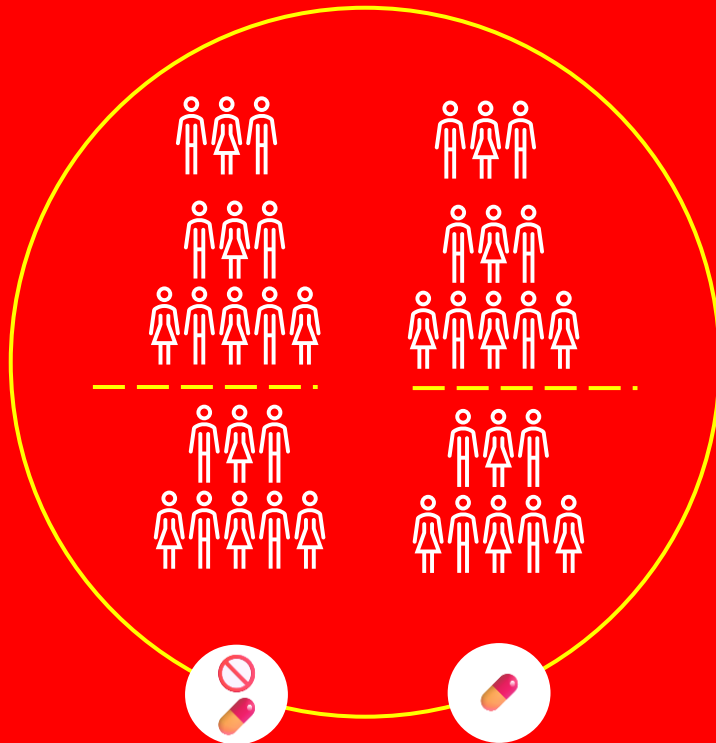
Within the population, a medication administered to reduce blood pressure **has no effect**.

$$H_0: (Y | \text{pill}) = (Y | \text{no pill})$$

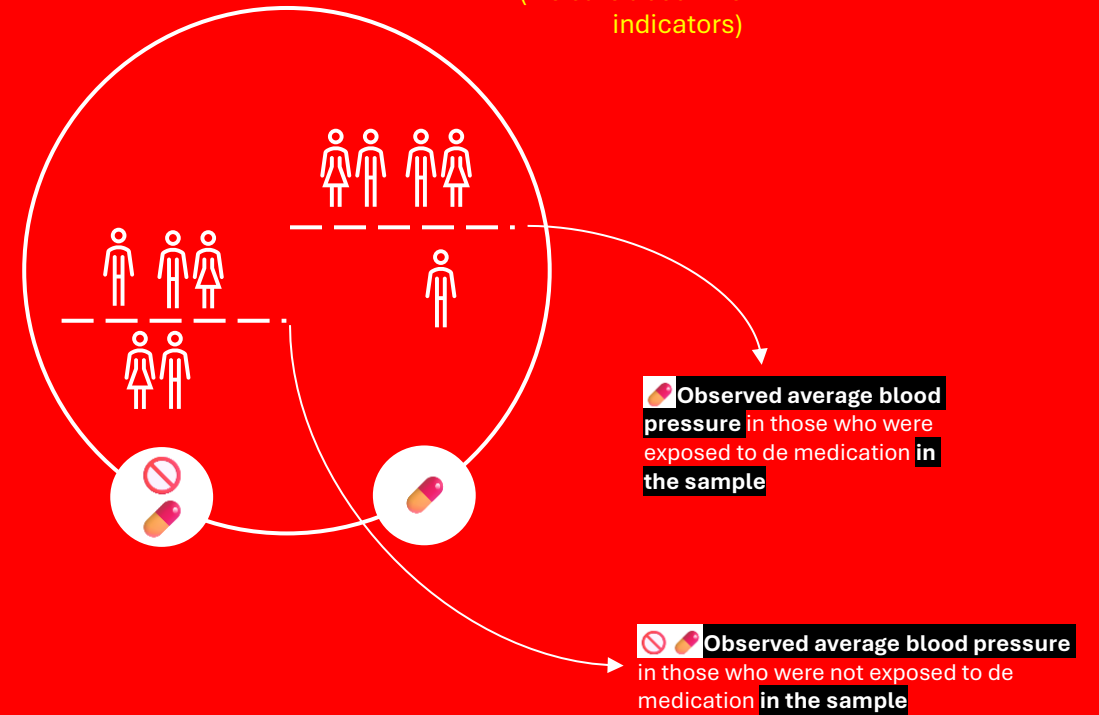
Under the null hypothesis, the average blood pressure level of individuals who took the medication () and those who did not take the medication () **is the same**.

Quantitative designs

Population under H0
(we care about the
parameters)

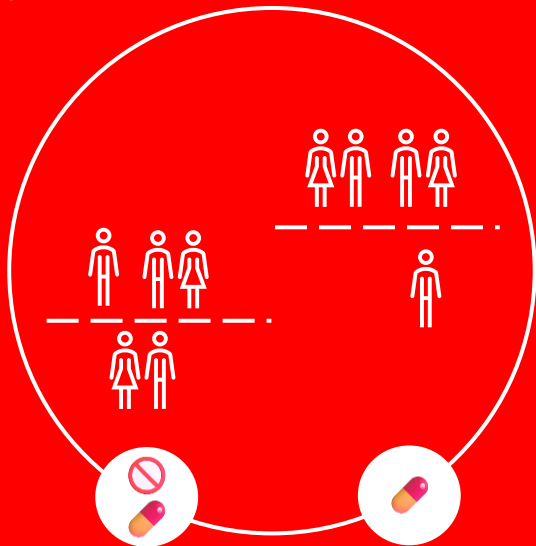


Observed sample
(we care about the
indicators)



Quantitative designs

Observed sample
(we care about the
indicators)



How probable is it to obtain a sample in which the estimated effect appears as least as large as observed, if the medication has no effect in the population?

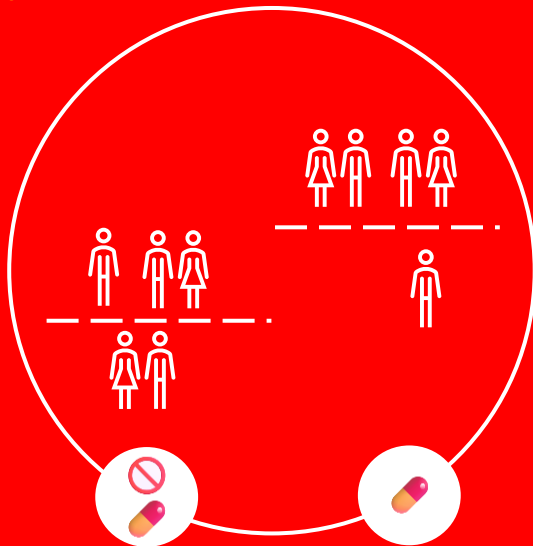
If this probability
is low:

Then we must ask whether, in the population, the medication might indeed have an effect.

In such cases, the observed sample result may provide sufficient evidence to reject H_0 , because what we observe in the sample is inconsistent with what is established under the null hypothesis (H_0).

Quantitative designs

Observed sample
(we care about the
indicators)



How far is the observed sample result from what the null hypothesis (H_0) predicts?

P-Value:

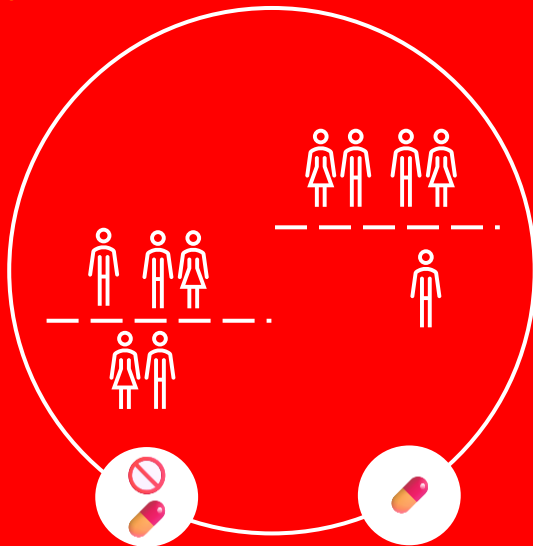
The p-value answers this question.

It is the probability of obtaining a sample whose indicators deviate as much as, or more than, those observed, **assuming that H_0 is true.**

If this probability is low, the sample result is inconsistent with H_0 , **providing grounds to reject the null hypothesis.**

Quantitative designs

Observed sample
(we care about the
indicators)



P-Value:

If the p-value is 0.01

- ✗ You are **NOT** saying “there is a 99% probability that the treatment works.”
- ✓ What you are saying is: “If the treatment did not work, I would see data like this in only 1 out of every 100 experiments. Since that would be very rare, I am more inclined to believe that the treatment does have an effect.”

Quantitative designs

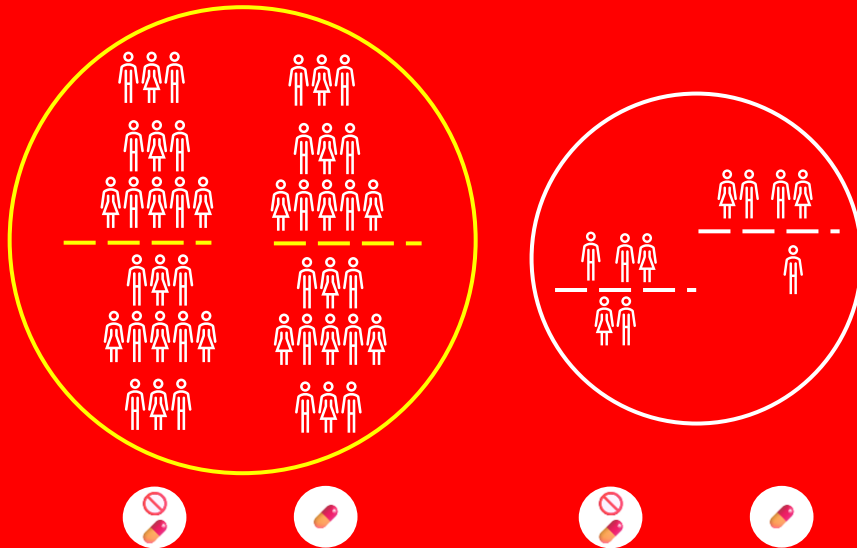
	H ₀ True	H ₀ False
Reject H ₀	✗ Type I error	✓ Correct decision
Do not reject H ₀	✓ Correct decision	✗ Type II error

Null Hypothesis (H₀)

Within the population, a medication administered to reduce blood pressure has no effect.

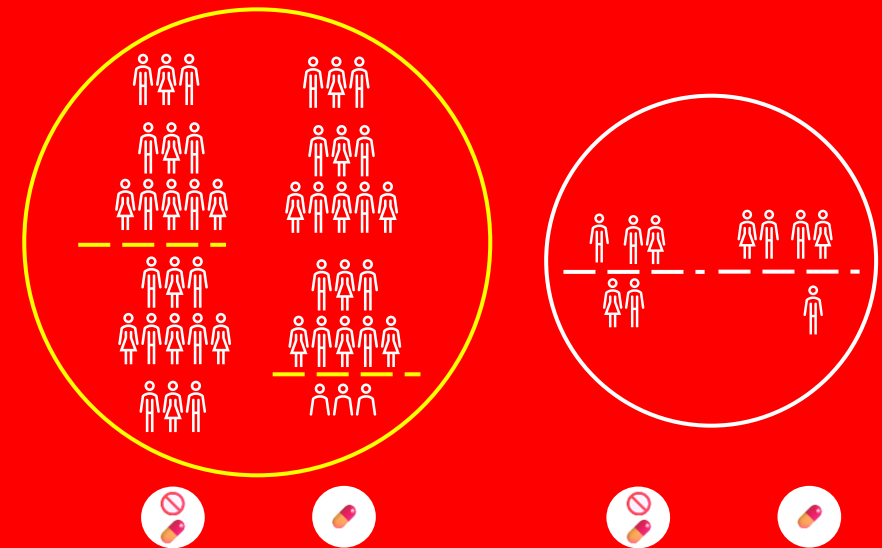
Type I error

Rejecting H₀ when it is in fact true.



Type II error

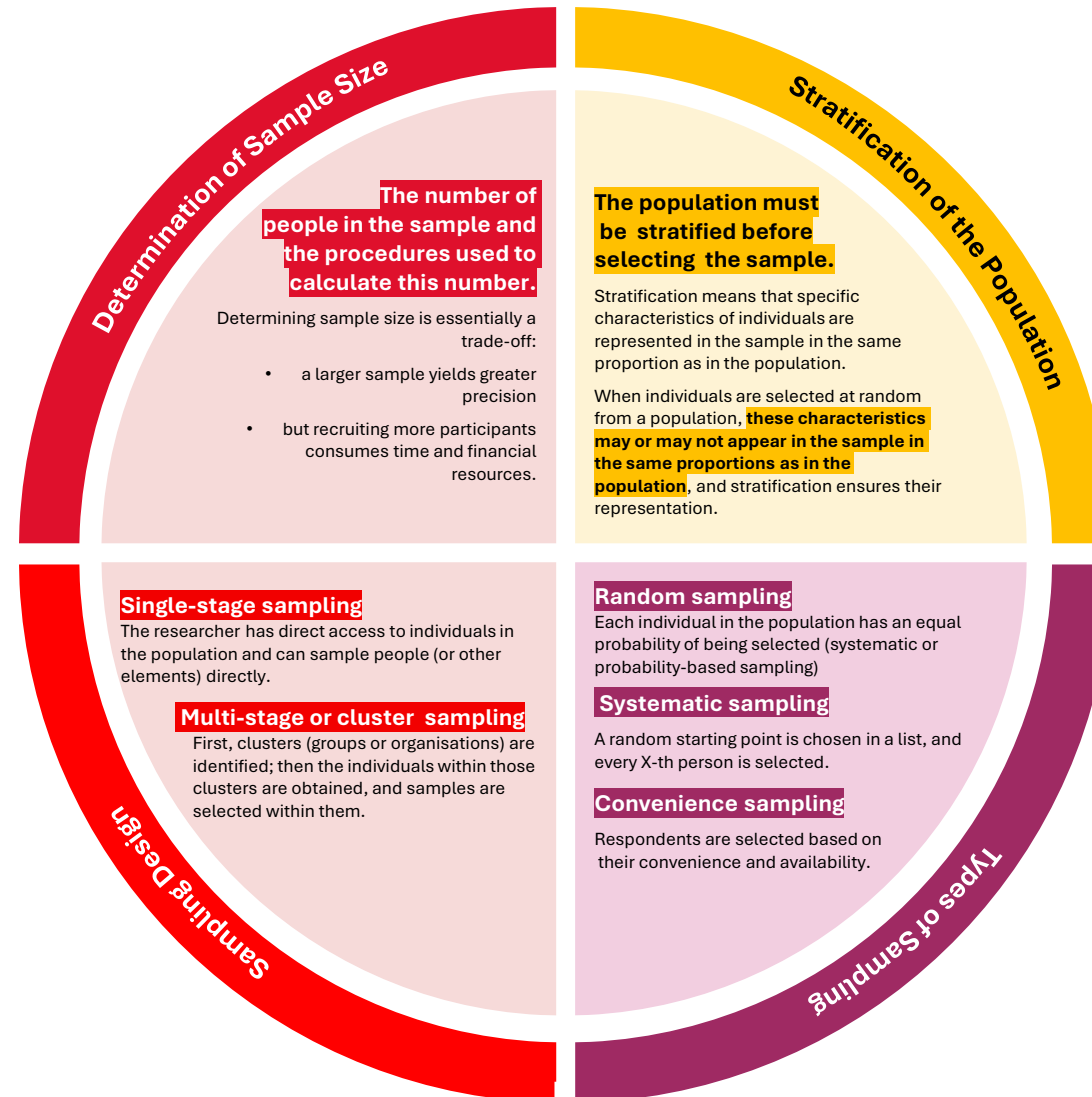
Failing to reject H₀ when it is in fact false.



Quantitative designs

Surveys

Sampling



Quantitative designs

Determination of Sample Size

In which scenario is your research situated?

- Sample sizes determined by **convenience**, due to operational, economic, or ethical constraints.
- The survey design (or experimental design) **must adapt to these limiting factors**, to the operational aspects of selecting the sample, and to the identification of the population.
- More complex survey or experimental designs **require larger sample sizes**.

1. Sample-to-item ratio

Relationship between sample size and number of items

- ✓ If your questionnaire contains 50 items, a 10:1 ratio suggests a sample of 500 participants.

2. Sample-to-variable ratio

Relationship between sample size and number of variables

- ✓ If your study includes 5 independent variables, a 20:1 ratio indicates a sample of 100 participants.

3. Krejcie and Morgan table

A table that provides the recommended sample size for different population sizes

- ✓ For a population of 1,000, the table suggests a sample size of 278 participants.

4. Roscoe, Green y Kline guides

Expert recommendations for determining adequate sample sizes in research.

- ✓ Roscoe suggests a minimum of 30 participants per group in experimental studies.

Quantitative designs

Determination of Sample Size: **Power**

Sample-size calculations (based on power) **depend on the specific hypothesis test** that will be used in the research:

Example 🧐

- ✓ Between-subjects design
- ✓ Null hypothesis of no treatment effect $H_0: \mu_1 = \mu_2$
- ✓ Analysis via linear regression or a t-test
- ✓ Continuous outcome variable.

- 
- 1** Significance level
 - 2** Test power
 - 3** Effect size

Quantitative designs

Determination of Sample Size: Power

1**Significance
level**

α is the probability of committing a Type I error, that is, rejecting the null hypothesis (H_0) when it is true.

This corresponds to a **false positive**: concluding that there is a treatment effect (or a correlation) when there is none.

Therefore, it is essential to choose an appropriate significance level for your tests to minimise the probability of such errors.

It is commonly set at 5%, although other levels may be used.

2**Test
power****3****Effect
size**

Quantitative designs

Determination of Sample Size: Power

2**Test
power**

Power of the test ($1 - \beta$)

Power is the probability of **correctly rejecting the null hypothesis** when the true effect is not zero (e.g., ATE $\neq 0$ or the correlation is non-zero).

A conventional benchmark is 80% power: β is the Type II error, failing to reject H_0 when it is false (a false negative).

An experimental design is said to have 80% statistical power if 80% of all possible random assignments would produce observable results (i.e., allow detection of the effect) when the effect actually exists.

1**Significance
level****3****Effect
size**

Quantitative designs

Determination of Sample Size: Power

3

Effect
size

Effect Size (relationship) and Minimum Detectable Effect (MDE)

(SES or d) The effect size is the magnitude of the relationship or difference that the study aims to detect. Planning requires specifying a minimum detectable effect (MDE).

- To compute it, one must first identify the minimum detectable effect (MDE).
- MDE: the minimum difference in outcomes between the treatment group (T) and the control group (C) that the study is designed to detect.

1

Significance
level

2

Test
power



(Cohen, 1969)

$$SES = MDE/SD = \frac{\bar{y}_t - \bar{y}_c}{\sigma_c}$$

Rules of thumb :

- If SES = 0.2, the effect size is small.
- If SES = 0.5, the effect size is medium.
- If SES = 0.8, the effect size is large.

In practice, the sizes of MDE, SD and SES are often taken from:

- Published studies with similar interventions or objectives
- Pilots.

Quantitative designs

Determination of Sample Size: Power

If your analysis plan includes **detecting a significant association between variables of interest**, a power analysis can help estimate a target sample size.

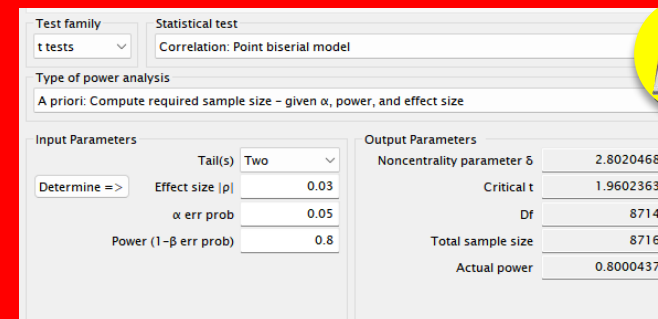
- The input values for a formal power analysis will **depend on the questions you wish to address in your design**.
- This power analysis for determining sample size must be **conducted during the planning phase of the study and before enrolling any participants**.

Hypothesis: there exists an association between hours of class and subjective well-being.

Estimated correlation (r): can be obtained from prior, similar studies.

Alpha (α) — significance: Type I error rate, commonly set at 0.05 (a 5% risk of a false positive).

Beta (β): Type II error rate, commonly set at 0.20 (80% statistical power — risk of a false negative).



The screenshot shows a software interface for power analysis. It includes a 'Test family' dropdown set to 't tests' and a 'Statistical test' dropdown set to 'Correlation: Point biserial model'. The 'Type of power analysis' is set to 'A priori: Compute required sample size - given α , power, and effect size'. The 'Input Parameters' section includes a 'Determine =>' button, 'Tail(s)' set to 'Two', 'Effect size |p|' set to 0.03, ' α err prob' set to 0.05, and 'Power (1 - β err prob)' set to 0.8. The 'Output Parameters' section displays calculated values: 'Noncentrality parameter δ ' (2.8020468), 'Critical t' (1.9602363), 'Df' (8714), 'Total sample size' (8716), and 'Actual power' (0.8000437). A circular logo with the Greek letters β and α is overlaid on the right side of the interface.

Quantitative designs

Determination of Sample Size: Power

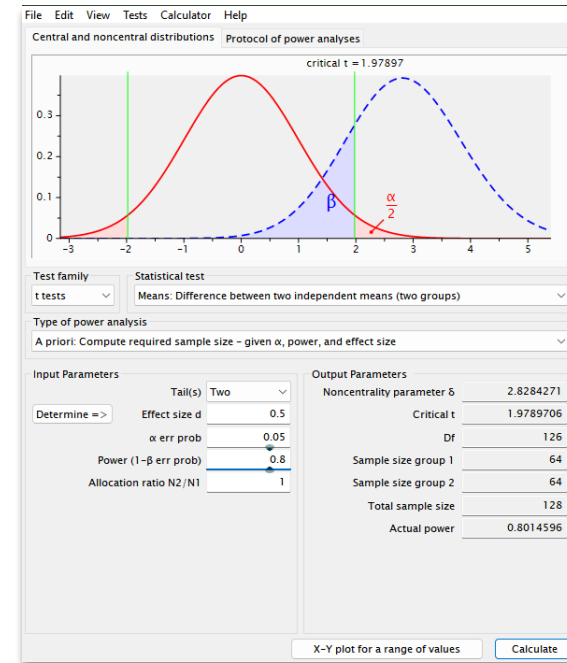
1. Define the test type

- Menu : Tests
 - Tests: Means
 - Medias: Two independent groups

The screenshot shows the G*Power software interface. The 'Test family' is set to 't tests' and the 'Statistical test' is 'Means: Difference between two independent means (two groups)'. The 'Type of power analysis' is 'A priori: Compute required sample size - given α , power, and effect size'. The 'Input Parameters' section shows 'Tail(s)' as 'One', 'Effect size d' as 0.5, ' α err prob' as 0.05, 'Power (1- β err prob)' as 0.95, and 'Allocation ratio N2/N1' as 1. The 'Output Parameters' section shows 'Noncentrality parameter δ ' as '?', 'Critical t' as '?', 'Df' as '?', 'Sample size group 1' as '?', 'Sample size group 2' as '?', 'Total sample size' as '?', and 'Actual power' as '?'. A 'Calculate' button is at the bottom right.

2. Define the parameters for the power calculation

- Effect size: Medium $\sim$ SES = 0.5
- Tests: Two tails $\sim$ given the structure of the Null $H_0: \mu_1 = \mu_0$
- Power: $(1 - \beta) = 0.8$
- Significance: $\alpha = 0.5$

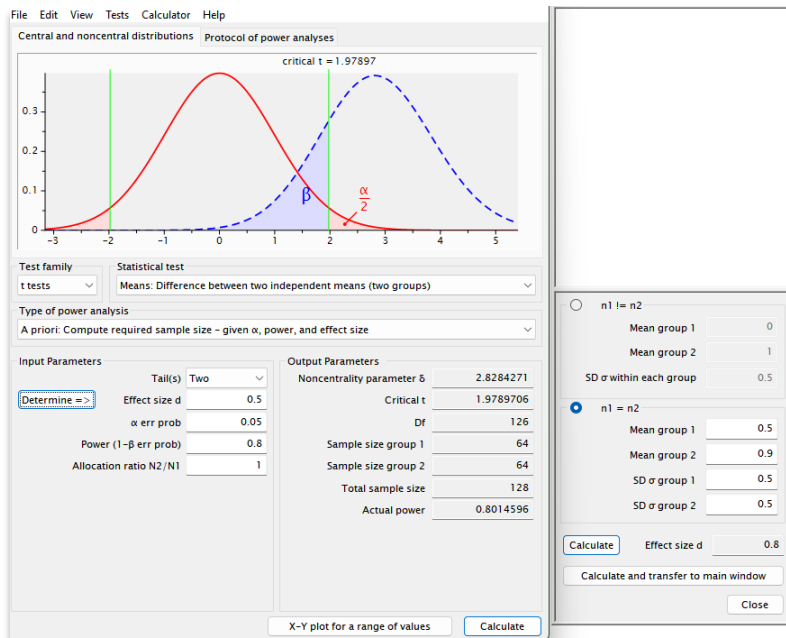


Quantitative designs

Determination of Sample Size: Power

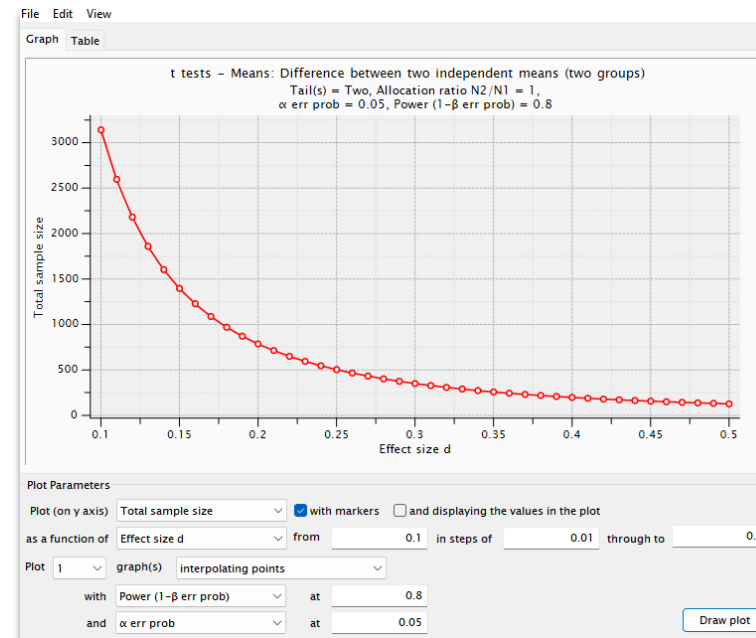
3. Simulate the effect size

- Input parameter : Determine
 - Medias: by group
 - Deviations: by group
 - Individuals per group: by case



4. Simulate sample size scenarios

- X Y plot for a range of values:
 - Power: $(1 - \beta)$
 - Significance: α
 - Effect size: SES/d



Quantitative designs

Determination of Sample Size: **Power**

1. Define the test type

- **Menu** : Tests
 - **Tests**: F tests
 - **Multiple linear regression**: Fixed model, R^2 increase

In the context of multiple linear regression, Cohen's f^2 **is a measure of effect size that evaluates the proportion of variance explained by a set of predictors relative to the unexplained variance**:

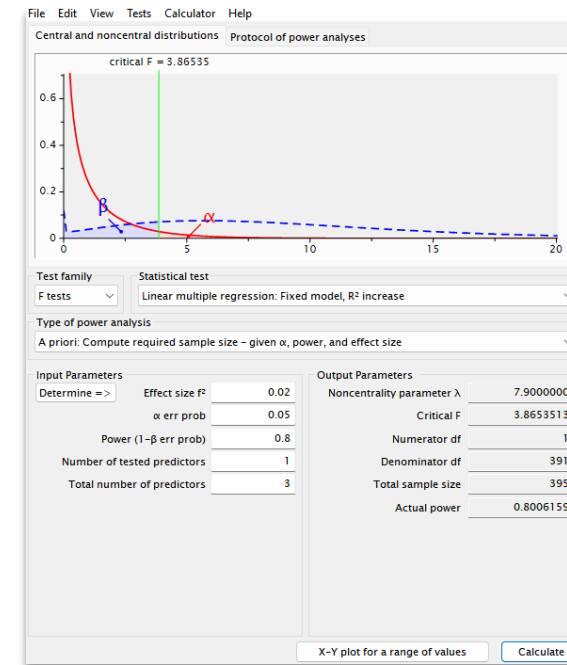
$$f^2 = \frac{R^2}{1-R^2}$$

- $f^2 = 0.02$: **small effect**
- $f^2 = 0.15$: **medium effect**
- $f^2 = 0.35$: **large effect**

When conducting a power analysis to calculate sample size, f^2 is a key input for determining **how many participants are needed to achieve adequate statistical power** (e.g., 80%).

2. Define the parameters for the power calculation

- **Number of predictors**: 3
- **Number of tested predictors**: 1
- **Power**: $(1 - \beta) = 0.8$
- **Significance**: $\alpha = 0.5$



Quantitative designs

Determination of Sample Size: Power

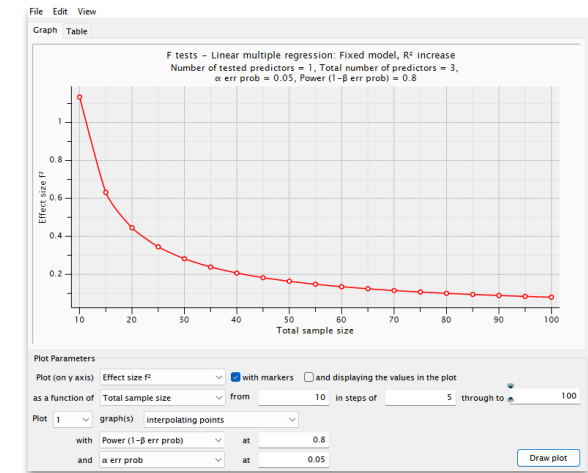
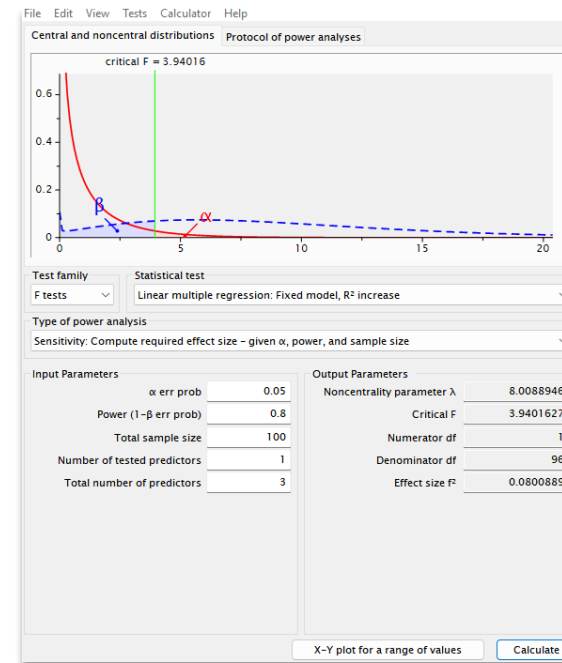
What happens if the number of participants is limited by the available resources?

Sometimes, limited resources (e.g., time, budget, or participant availability) **determine the maximum number of participants you can include in your study.**

- ✓ **Sensitivity and power analysis is used to compute the minimum effect size that can be reliably detected with the available sample size, while maintaining, for example, 80% power and a significance level of $\alpha = 0.05$.**

Define the parameters for the power calculation

- Number of predictors: 3
- Number of tested predictors: 1
- Sample : 100
- Power: $(1 - \beta) = 0.8$
- Significance: $\alpha = 0.05$



Quantitative designs

Surveys

1

Purpose of the
Instrument

2

Sampling
Process

3

**Instrument
Development**

Instrument Development

In a survey, questions and answers are **relevant because of their relationship to what they are supposed to measure**.

Good questions are:

- ✓ **Reliable** (they provide consistent measurements in comparable situations)
- ✓ **Valid** (the answers correspond to what they are intended to measure)

Designing a survey question means **designing a measurement, not engaging in a conversational inquiry**.

- The value of the response depends on whether it can be shown to have a predictable relationship with the facts or subjective states of interest.

Good questions maximise the relationship between the **recorded answers** and what the researcher **seeks to measure**.

Survey responses **are reactions evoked in an artificial situation** designed by the researcher.



[Fowler \(2014\)](#)

A respondent tells us that they voted for Rodolfo instead of Petro in the 2022 presidential election.



We care about is **which box they marked in the voting booth**.

The response given in the **survey may differ from what truly happened** for several reasons:

- ✗ The respondent may have **ticked the wrong box** and therefore may not actually know who they voted for.
- ✗ The respondent may have **forgotten** who they voted for.
- ✗ The respondent may have **intentionally altered their answer** for some reason.

A respondent tells us how many times they visited a doctor for medical care in the past year

- ✗ Would this number be the same as what the researcher would calculate if they **followed the respondent 24 hours a day over the past year**, or checked administrative records?
- ✗ Memory problems, uncertainty about **what counts as a doctor's visit**, and willingness to report accurately may affect the correspondence between the respondent's answer and the count the researcher would independently obtain.

Instrument Development

In a survey, questions and answers are **relevant because of their relationship to what they are supposed to measure**.

Good questions are:

- ✓ **Reliable** (they provide consistent measurements in comparable situations)
- ✓ **Valid** (the answers correspond to what they are intended to measure)

Designing a survey question means **designing a measurement, not engaging in a conversational inquiry**.

- The value of the response depends on whether it can be shown to have a predictable relationship with the facts or subjective states of interest.

Good questions maximise the relationship between the **recorded answers** and what the researcher **seeks to measure**.

Survey responses **are reactions evoked in an artificial situation** designed by the researcher.



Fowler (2014)

Many surveys are analyzed and interpreted as if the researcher knew with certainty what the respondent's answer means.

- ✗ It is strongly recommended **NOT to assume a perfect correspondence between respondents' answers and the real phenomena those answers are supposed to represent**

1. Reliability

When two respondents are in the same situation, **they should respond to the question in the same way**.

Whenever inconsistency arises among respondents, **random error is introduced and the measurement becomes less precise**.

2. Validity

Validity refers **to what a response means in relation to what the researcher intends to measure**, that is, the correspondence between the response and the true value, and how this correspondence can be improved.

Instrument Development

Reliability

In an ideal setting, it is assumed **that differences in respondents' answers** are due to differences in what respondents have to say, and **not due to differences in the stimuli to which they have been exposed**.

Data/information collection

Data collection in a survey is an **interaction between a researcher and a respondent**.

- ✓ **In a self-administered survey**, whether on paper or digitally, the researcher communicates directly with the respondent through the written questionnaire or the words on the device screen
- ✓ **In interviewer-administered surveys**, the interviewer reads the researcher's scripted words to the respondent.

In both cases, the survey instrument is the protocol governing part of the interaction

For the data collection experience to be **consistent for all respondents**, a **good question** must have the following properties:

- 1.** The types of responses that constitute an adequate answer to the question are communicated consistently to all respondents.
- 2.** The question means the same thing for all respondents.
- 3.** The question-and-answer process is fully scripted, so that the questions, exactly as written, fully prepare the respondent to answer.

Instrument Development

Reliability

Inadequate Wording

Wording does not constitute a complete question.

Age?



How old were you on your last birthday?



Reason for going to the doctor?



What was the main medical problem or reason for which you most recently visited the doctor?



Interviewers (or respondents) may **need to add or modify words** in order to answer the questions based on the wording provided in the left-hand column.

Optional Wording

At times, optional wording is necessary to adapt to respondents' circumstances.

In virtual interviews, it is possible to **design multiple versions of the questions**.

In paper questionnaires, **a common convention is to place optional text in parentheses**.

- ?** Were you (or anyone living with you) assaulted or struck by a stranger during the past year?
- ?** Did (you/he/she) report the assault to the police?
- ?** How old was (EACH PERSON) on their last birthday?

Instrument Development

Reliability

Unacceptable Phrasing

Wording does not constitute a complete question.

What do you like most about this neighborhood? (We are interested in anything such as the houses, the people, the parks, or whatever it may be.)

From a measurement perspective, the use of clarifying text in parentheses **undermines the principle of the standardised interview.**

- ✗ If interviewers rely on parenthetical questions when a respondent does not answer easily, **that subset of respondents will have effectively answered a different question.**
- ✗ **Interviewers must not be given discretion regarding which questions to read or how to read them**, except where wording must be standardly adapted to a respondent's specific circumstances.

The correct procedure is to **formulate a well-designed question from the outset.**

Incomplete phrasing

I would like you to rate different characteristics of your neighborhood as very good, good, fair, or poor. Please think carefully about each item as I read them.

- Public schools • Public transport
- Parks • Others

- ✗ The response options are read before the instruction to reflect carefully on the specific items.
- ✗ The respondent is likely to forget the question.
- ✗ The interviewer will likely need to explain or rephrase the question before the respondent is prepared to answer.

I am going to ask you to evaluate different characteristics of your neighborhood. I want you to think carefully about your responses. How would you rate (CHARACTERISTIC)?

- good, fair, or poor

Instrument Development

Reliability

Poorly Defined Items

Questions must mean the same thing for all respondents. If two respondents interpret the meaning of the question differently, their answers **may differ solely for that reason**, rather than because of substantive differences in their experiences or views.

- ✗ **Use of words that are not universally understood.** In general population samples, a wide variety of educational backgrounds and cultural contexts will be represented. Even among highly educated respondents, **using simple, short, and easily understood words is a sound approach to questionnaire design.**

- ✗ **Use of terms or concepts that may have multiple meanings.**

How many times in the last year have you visited or spoken to a doctor about your health?



- ✗ **What counts as a doctor?** Does it include only those with a medical degree? Are psychiatrists included? Are psychologists, chiropractors, osteopaths, or podiatrists excluded?
- ✗ **What counts as seeing or speaking to a doctor?** Does a nurse's visit count? Does email communication count?
- ✗ If an individual goes to a doctor's office for a nurse to administer a vaccine, does that count?

We would like to ask you about your visits to a doctor and the medical advice you have received. In this question, we are interested in all professional staff who hold a medical degree or who work directly for a doctor in the practice, such as nurses or physician assistants



An additional approach is to include follow-up questions about events that respondents are especially likely to omit.

Instrument Development

Reliability

Poorly Defined Items

Did you have breakfast yesterday?



- ✗ **The definition of breakfast varies considerably across respondents.**
- Coffee and a pastry at any time before noon
 - It MUST include a main dish such as eggs, eaten early in the morning.

If the objective is to **measure morning food consumption**, the results will likely contain error due to inconsistent interpretations of what “breakfast” means.

For our purposes, consider breakfast to be a meal eaten before 10:00 a.m. that includes a protein (such as eggs, meat, or milk), a cereal (such as toast or cereal), and a fruit or vegetable, including juice. According to this definition, did you have breakfast yesterday?



- ✓ This approach provides a shared definition that standardizes the measurement.

Instead of attempting to impose a definition on respondents, the researcher **may simply ask respondents to report everything they consumed before 10:00 a.m.**

During coding, the researcher **can then evaluate objectively whether the respondent's food intake meets the established definition of breakfast**, without requiring all respondents to share the same understanding of the term.

Instrument Development

Reliability

Poorly Defined Items

Are you in favour of or against legislation on front-of-package nutritional labelling?



Legislation on food-labelling warnings may involve:

- Requiring producers to label their food products clearly and accurately.
- Setting limits on the permitted levels of components such as sugars, fats, or sodium in food products.
- Establishing restrictions on the types of products that may be marketed to the public.

Consumers' **responses cannot be interpreted without making assumptions about how they understand the meaning of the labelling.**

A proposal for food-control policy is that no food product exceeding certain permitted levels of sugar, fat, or sodium may be sold without a clear nutritional warning label. Would you be in favour of or against legislation of this kind?



This is only **one of several possible strategies** for regulating food products through nutritional warning labels.

That is precisely the point:

if multiple questions about different regulatory strategies are required, they should be separate, specific, and phrased in a way that all respondents can understand uniformly and that researchers can interpret correctly.

Instrument Development

Reliability

Poorly Defined Items

Do you want to be rich and famous?



In the past 30 days, when you withdrew money from an ATM, how often was it less than 100,000 pesos: always, usually, sometimes, never?



In the past 30 days, how many times did you withdraw money from an ATM



(IF APPLICABLE) In how many of those times did you withdraw less than 100,000 pesos?



Some questions that appear to be single items **actually require the respondent to answer several sub-questions before producing a response.**

- ❓ The number of ATM visits
- ❓ The number of times they withdrew less than 100,000 pesos
- ❓ The relationship between the two quantities

Although technically one question, **it requires multiple mental calculations,** compromising reliability.

Instrument Development

Reliability

Response Options: “Don’t know”

When respondents are asked about their own lives, feelings, or experiences, a “don’t know” response is often an indication that **they are not willing to invest the cognitive effort required to provide an answer.**

- ✓ Offering the option “**no opinion**” tends to improve measurement by inducing a greater number of respondents to answer.

However, in some cases, **respondents are legitimately unaware of the topic.** The further the topic is from their immediate lives, the more plausible it becomes that respondents lack the information needed.

- ✓ When a notable proportion of respondents lacks the necessary information to answer a factual question, it is often best to **include a filter (screening question)**

In such cases, the question effectively contains two questions at once:

- 1. Do you have the information necessary to answer this question?**
- 2. If so, what is your answer?**

Instrument Development

Reliability

Standardised expectations for the type of response

In addition to:

- ✓ Providing interviewers with a good script so that they can read the questions exactly as written
- ✓ Designing questions that mean the same thing for all respondents

Another key component of a good question is ensuring that **respondents share the same perception of what constitutes an adequate answer.**

Closed questions

The simplest way to ensure that respondents share a common understanding of what counts as an acceptable response is to **provide a list of acceptable answers.**



What was the main reason you visited the doctor? A new health problem, a follow-up for a previous problem, a routine check-up, or any other reason?

Closed questions **are not suitable in all cases.**

- ✗ The range of possible responses may be too broad to list reasonably.
- ✗ The researcher may believe that not all reasonable answers can be anticipated in advance.

In such cases, the question **should communicate as clearly as possible the type of response that is desired.**

Instrument Development

Reliability

Standardised expectations for the type of response

When did you have pneumonia?



The question does not specify the terms in which the respondent is expected to answer.

All of the following could be provided by the same person, and all are “correct” answers given the way the question is written:

- ✗ “Five years ago”
- ✗ “While I was in the military”
- ✗ “When I was pregnant with our first child”
- ✗ “When I was 32”
- ✗ “In 1987”

Because these responses differ in format and meaning, the lack of standardised expectations reduces reliability.

“How old were you when you had pneumonia?”



Instrument Development

Validity

Question types

You can obtain **perfect reliability and still not be measuring what you intend to measure.**

The extent to which the response **provided is a true measure and means what the researcher wants or expects it to mean** is called validity.

Validity

To study validity, it is necessary to distinguish between:

- Questions designed to measure **objectively measurable facts or events**
- Questions designed to measure **subjective states** (such as attitudes, opinions, and feelings).

For the data collection experience to be **consistent for all respondents**, a **good question** must have the following properties:

Independent observation

The measure of validity becomes the **similarity between the survey report and the value of some “true” measure.**

Non available independent observation

The validity of reports about a person’s degree of happiness, tiredness, or interest **can only be assessed by their correlation with other responses** or with other facts from the respondent’s life that one believes ought to be related to what is being measured.

Instrument Development

Validity

Levels of measurement

The levels of measurement produce four different kinds of tasks for respondents and **four different kinds of data for analysis**.

1. Nominal

People or events are classified into unordered categories.

(Are you male or female?)

2. Ordinal

People or events are ordered or placed into ordered categories along a single dimension.

(How would you rate your health: very good, good, fair, or poor?)

3. Interval

Numbers are attached that provide meaningful information about the distance between ordered stimuli or classes.

(Fahrenheit temperature)

4. Ratios

Numbers are assigned so that ratios between values are meaningful, as are the intervals between them. Common examples are counts or measurements based on an objective physical metric, such as distance, weight, or time.

(How old were you on your last birthday?)

Instrument Development

Validity

Open and closed questions

What health problems do you have? (**open**)

Which of the following illnesses do you currently have? (READ THE LIST) (**closed**)

What do you consider to be the most important problem facing our country today? (**open**)

This is a list of problems that concern many people in the country. Which do you consider to be the most important problem facing our country today? (**closed**)

When the objective is **to classify people into unordered categories** (nominal data) **or ordered ones** (ordinal data), the researcher can choose to formulate either an open or a closed question.

1. Open questions

- ✓ **They allow** the researcher to obtain unexpected responses that can enrich the findings. They can also more accurately reflect respondents' true opinions.
- ✓ **They are useful** when the list of possible responses is too long to present in a practical way.

2. Closed questions

- ✓ **They make it easier** for respondents to answer reliably because the response options are defined.
- ✓ **They allow** researchers to interpret answers more precisely.
- ✓ **They avoid** uncommon responses that may not be analytically useful, ensuring that more respondents choose relevant options.

Instrument Development

Validity

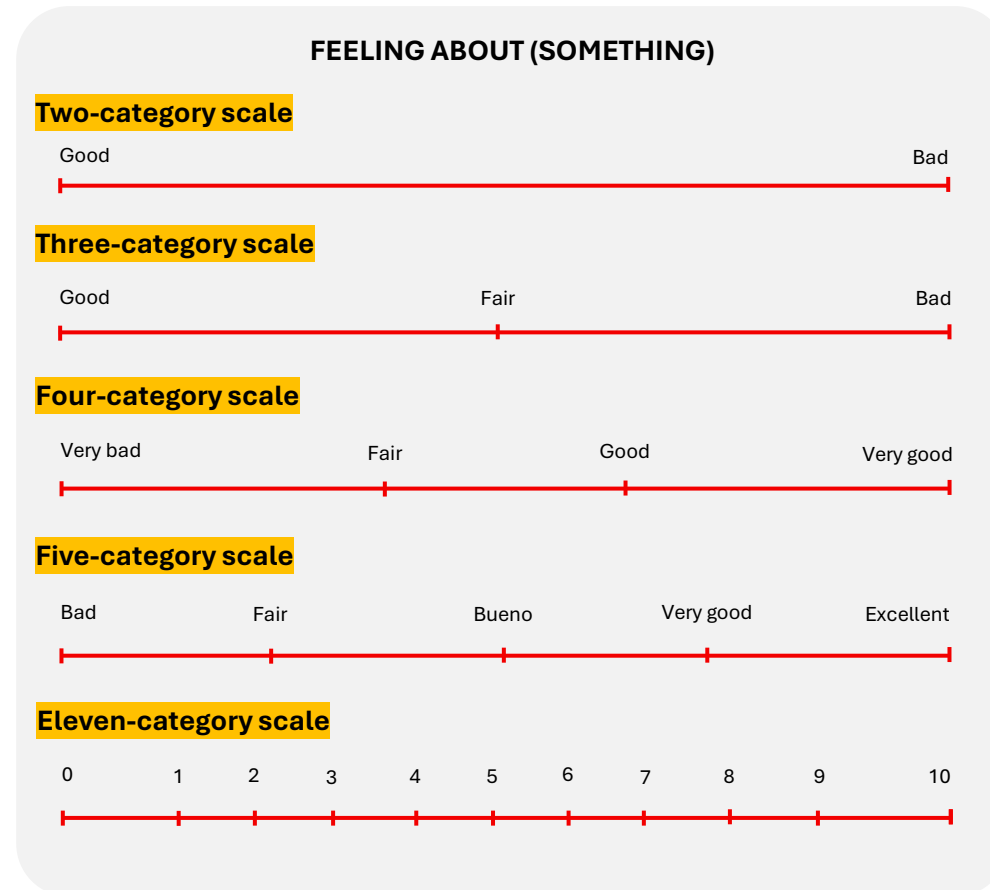
Types of questions : Continuous

If ordinal data are desired, **categories must be provided to respondents**.

One cannot reliably order responses along a single continuum unless the question specifies an admissible set of ordered responses.

There is an **assumed dimension** that runs from the most negative possible feelings to the most positive possible feelings.

- ✓ Place labels on a continuum
- ✓ Ask respondents to consider the labels, their own feelings or opinions, and to position the categories in the correct order.



Instrument Development

Validity

Types of questions : Agreement/Disagreement Items

The task presented to respondents in this type of item is **different from that of placing themselves in an ordered category**.

- ✓ Read a statement to respondents and ask whether they agree or disagree with it.

The statement is positioned along a continuum, and the respondent's position on that continuum is inferred from whether they are close to the statement (**agree**) or far from it (**disagree**).

"My health is poor. Do you strongly agree, agree, disagree, or strongly disagree?"

"How would you rate your health: excellent, very good, good, fair, or poor?"

Rating scales provide more information by classifying respondents into multiple categories, **whereas agreement-disagreement questions generally separate them into just two groups (agree/disagree)**.

- ✗ They can only address the extremes of a continuum, which limits the interpretation of intermediate values.
- ✗ Respondents may disagree with "My health is fair" because they consider it "good" or "poor."
- ✗ Respondents may find it confusing to express a positive state (for example, "good health") by disagreeing with a negative statement (for example, "poor health").

Acquiescence bias

Some respondents tend to agree with statements regardless of their content. They may agree both with "health is poor" and "health is not poor" if the statement is phrased negatively.

Instrument Development

Validity

Objective Reports

When a researcher asks a respondent an objective question, **the aim is for the respondent to answer with complete accuracy: to give the same answer the researcher would provide if the researcher had access to the necessary information to answer the question.**

The accuracy of responses depends on what is being asked and how it is being asked. There are four reasons why respondents may report events inaccurately:

- ✗ They do not understand the question.
- ✗ They do not know the answer.
- ✗ They cannot remember the answer, even if they once knew it.
- ✗ They do not want to provide the answer in the context of the interview.

1. They do not understand the question

If respondents do not all interpret what is being asked in the same way, error can arise. To avoid this, a researcher may:

- ✓ Provide definitions to all respondents.
- ✓ Ask respondents to provide the information needed to classify their experiences into detailed categories.

However, the first step in maximising the validity of objective survey data is **to write questions that all respondents can understand.**

2. They do not know the answer

A lack of knowledge as a source of error arises in two main ways

- ✗ The individual does not know the answer to the question, but someone else in the selected household does.
- ✗ No one in the selected household knows the answer.

In the first case, the solution lies in choosing the appropriate respondent, not in redesigning the question. Thus, it is possible to:

- ✓ Identify and interview the best-informed member of the household.
- ✓ Eliminate proxy respondents; ask only respondents to provide information about themselves.

Instrument Development

Validity

Objective Reports

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- ✗ They do not want to provide the answer in the context of the interview.

3. They cannot remember, even though they do know it

When respondents are asked questions about themselves that they cannot answer, this is a problem of question design.

In theory, **one could distinguish between information that the respondent cannot remember and information that the respondent never had.**

In both cases, the problem for the researcher is to design questions that almost everyone can answer. Possible solutions include:

- ✓ Rewriting the question to request information that is less detailed or easier to remember.
 - ✓ Helping the respondent estimate the answer.
 - ✓ Changing or removing the objective of the question.
- ✗ A question that asks respondents to list **the names of all the medications they take** (a very difficult question) when the objective is simply to find out **who is taking medication for hypertension** (an easier question).

Instrument Development

Validity

Objective Reports

When a researcher asks a respondent an objective question, **the aim is for the respondent to answer with complete accuracy: to give the same answer the researcher would provide if the researcher had access to the necessary information to answer the question.**

The accuracy of responses depends on what is being asked and how it is being asked. There are four reasons why respondents may report events inaccurately:

- ✗ They do not understand the question.
- ✗ They do not know the answer.
- ✗ They cannot remember the answer, even if they once knew it.
- ✗ They do not want to provide the answer in the context of the interview.

4. They do not want to give the answer in the context of the interview

It is important to recognise that there are some things that researchers would like people to report, **but they cannot.**

There are certain **facts or events that respondents would prefer not to report** accurately in an interview.

If a researcher realistically expects someone to admit something that is very embarrassing or illegal, **efforts are required to convince respondents that the risks are minimal and that the reasons for disclosing the information are meaningful.**

✓ Use of self-administered data-collection procedures

Such as paper or computer-based forms, **which reduce social desirability bias.**

✓ Confidentiality and anonymity

Ensure that responses **cannot be linked to respondents** through methods such as anonymous forms or other strategies specifically designed for sensitive topics.

✓ Minimising the sense of judgement

Avoid implicit judgements in the wording and **create a neutral response context**, since respondents interpret both the wording of questions and the interviewer's behaviour as cues about how their answers will be evaluated.

Instrument Development

Validity

Objective Reports

When a researcher asks a respondent an objective question, **the aim is for the respondent to answer with complete accuracy: to give the same answer the researcher would provide if the researcher had access to the necessary information to answer the question.**

The accuracy of responses depends on **what is being asked and how it is being asked**. There are four reasons why respondents may report events inaccurately:

- ✗ They do not understand the question.
- ✗ They do not know the answer.
- ✗ They cannot remember the answer, even if they once knew it.
- ✗ They do not want to provide the answer in the context of the interview.

Subjective Reports

The validity of subjective questions differs from that of objective questions because no external criterion exists for measuring it.

The only way to estimate the validity of a subjective measure is through the **expected association between responses to these questions and other characteristics of the individual**, or related responses.

This means that subjective measures are evaluated by patterns, not by universal standards. To increase the validity of these questions, it is essential that they be as reliable as possible. This includes:

- ✓ Eliminating ambiguities in wording
- ✓ Standardizing their presentation
- ✓ Avoiding imprecise response options

Instrument Development

Validity

1. **Defining the construct:** having a clear definition of what you want to measure.
2. **Developing theory-based questions:** formulating questions that align with the theoretical construct and reviewing the literature to ensure that all of its dimensions are covered.
3. **Involving experts:** to validate the content, invite experts to review the instrument.
4. **Collecting pilot data:** Conduct a pilot study with a small sample to test the instrument and run statistical analyses.
5. **Applying statistical analyses:**
 - For construct validity, use factorial analyses.
 - For criterion validity, use correlations with other instruments.
 - For nomological validity, test theoretical relationships with other constructs.